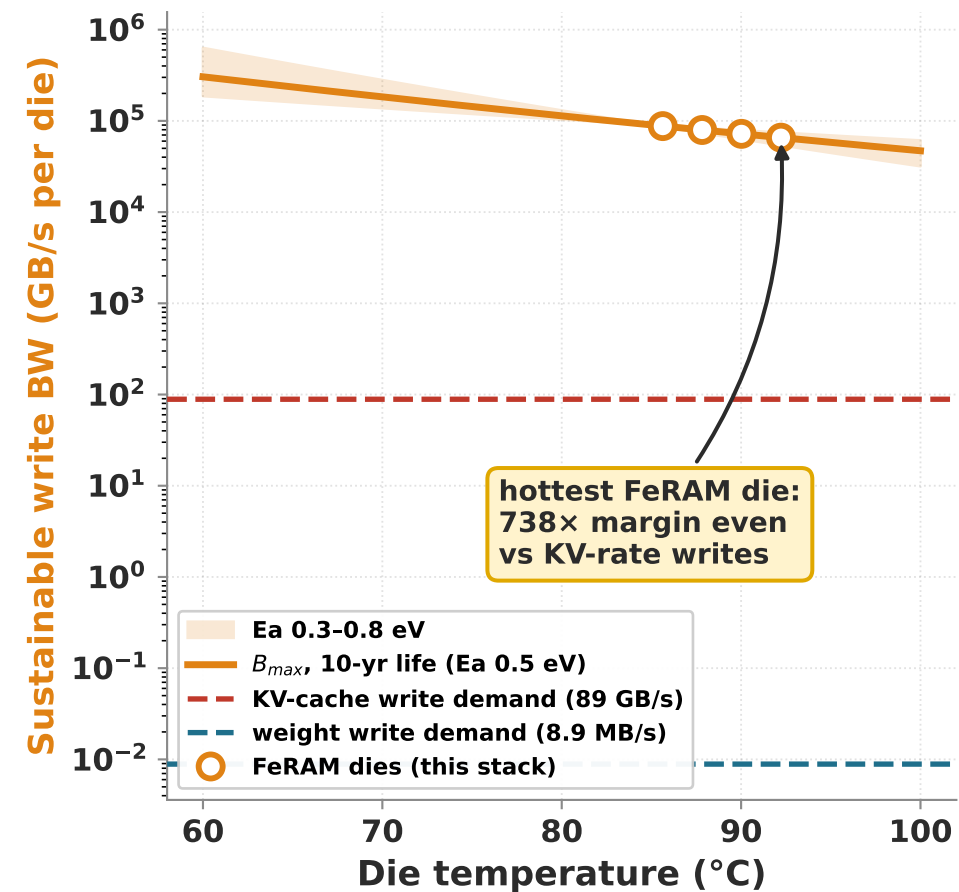
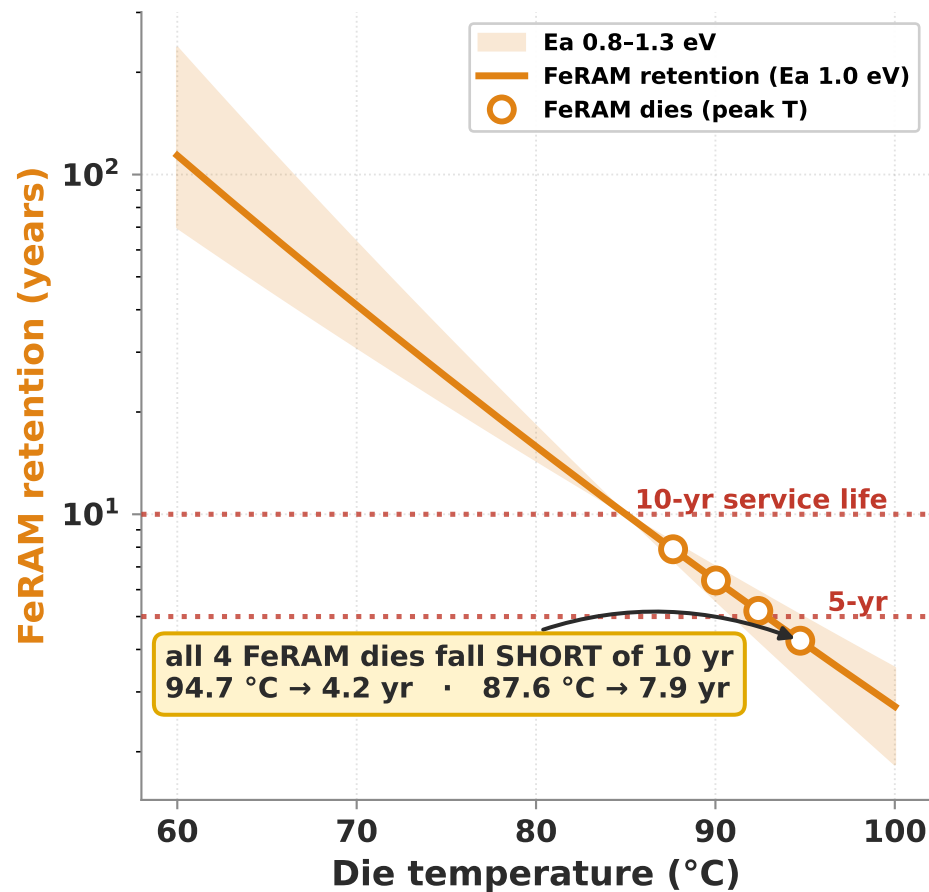


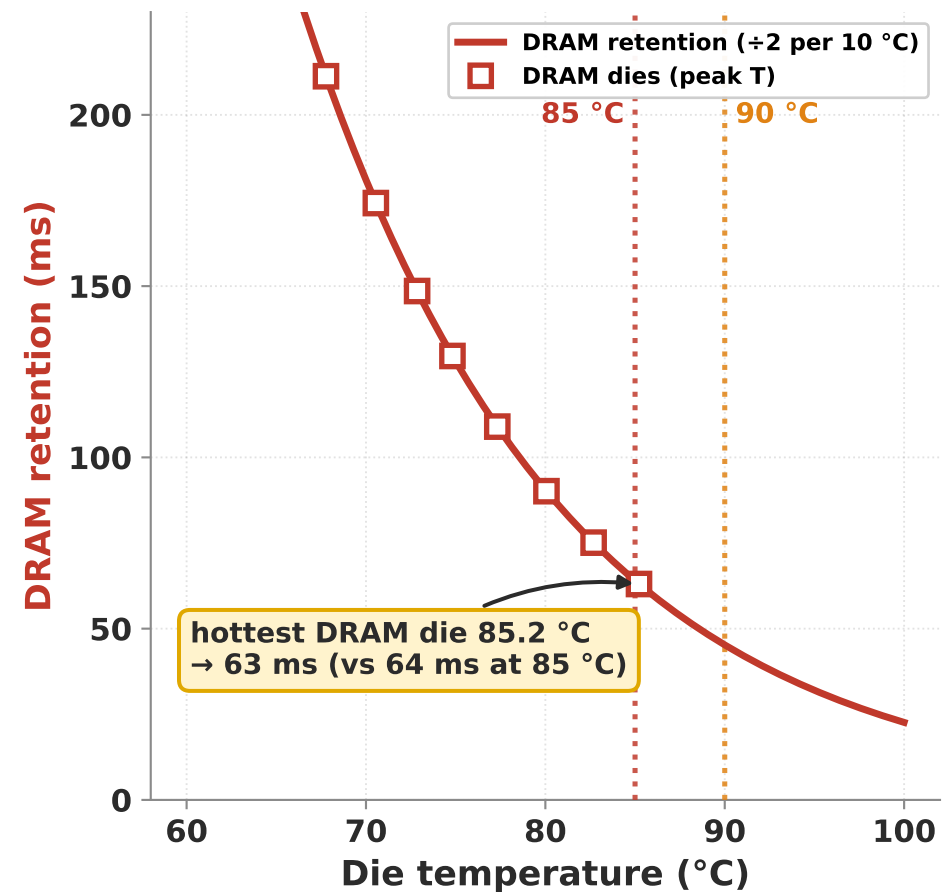
(a) Write endurance — huge margin



(b) FeRAM retention — the real limit



(c) DRAM retention — origin of the limits



Endurance: $N_0 = 1e+12$ cycles @ 85 °C, Ea 0.3-0.8 eV, wear-levelled over 28.5 GB/die (0.45 Gb/mm² × 506 mm²); traffic 4.4 TB/s/stack × $\alpha_w=0.2421$, $F_{write} = 0.0001$ (weights) / 1 (KV).

Endurance uses the wear-levelled ensemble temperature T_{eff} ; retention uses peak T (the hottest cell loses data first). Temperatures measured from PACT; reliability modelled.